



Visualizing Marketing Analytics of iFood Customer Data Using an Interactive Dashboard

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Introduction to the Dataset

- Dataset: Marketing Analytics of iFood Customer Data
- Consists of 2205 data instances (rows) and 39 attributes (columns)
- Contains data such as **customer demographics, spending behavior of customers, purchase channels, campaign responses, and customer activity**
- The dataset had undergone partial preprocessing before analysis, including
 - One-hot encoding of categorical variables
 - Generation of aggregated customer spending and campaign-response metrics

Dataset Overview

Customer Demographics	Household Composition	Customer Relationship & Activity	Product Spending Behavior	Purchase Channel Behavior
Income	kidhome	recency	mnt_wines	num_web_purchases
Age	teenhome	customer_days	mnt_fruits	num_catalog_purchases
			mnt_meat_products	num_store_purchases
			mnt_fish_products	num_web_visits_month
			mnt_sweet_products	num_deals_purchases
			mnt_gold_prods	

Dataset Overview

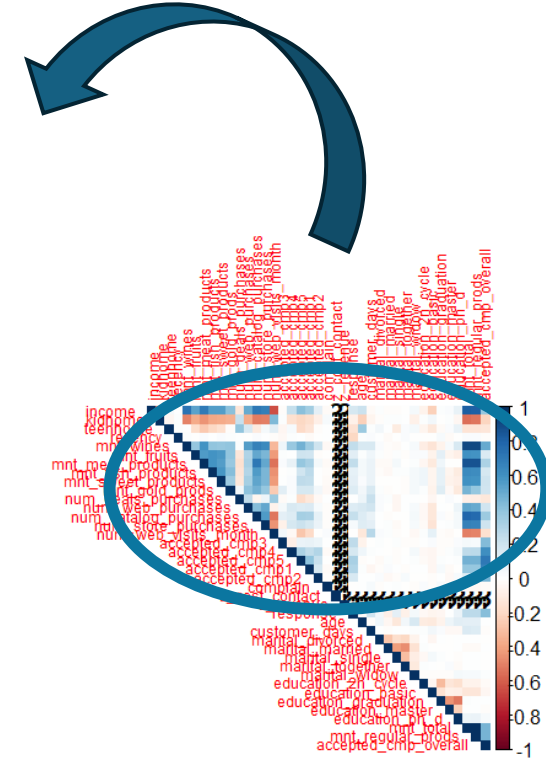
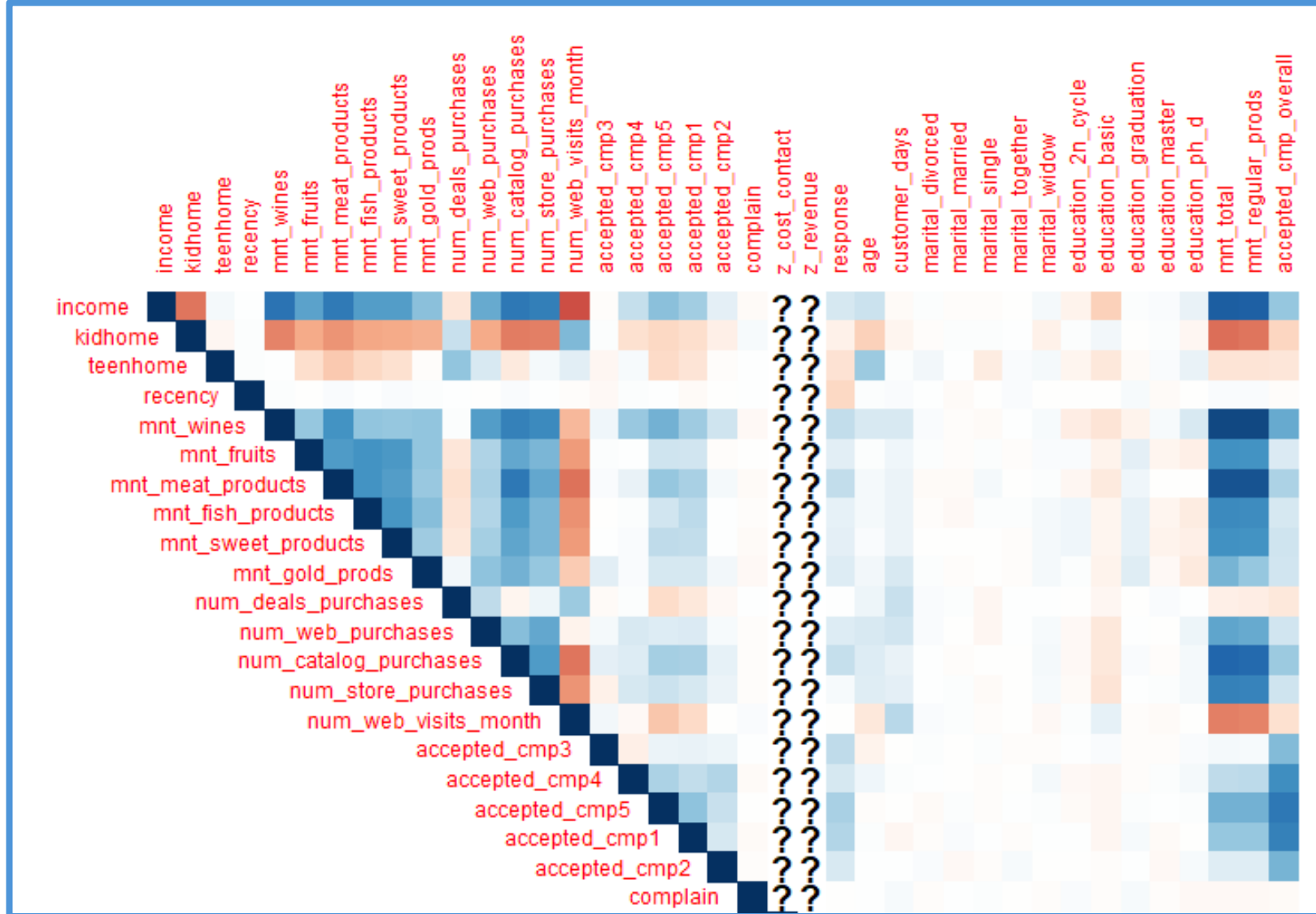
Marketing Campaign Responses	Customer Feedback & Complaints	Cost & Revenue Variables	Marital Status Features (One-Hot Encoded)	Education Features (One-Hot Encoded)	Engineered / Derived Features
accepted_cmp1	complain	z_cost_contact	marital_divorced	education_2n_cycle	mnt_total
accepted_cmp2		z_revenue	marital_married	education_basic	mnt_regular_prods
accepted_cmp3			marital_single	education_graduation	accepted_cmp_ove rall
accepted_cmp4			marital_together	education_master	
accepted_cmp5			marital_widow	education_ph_d	
response					

Important Observations

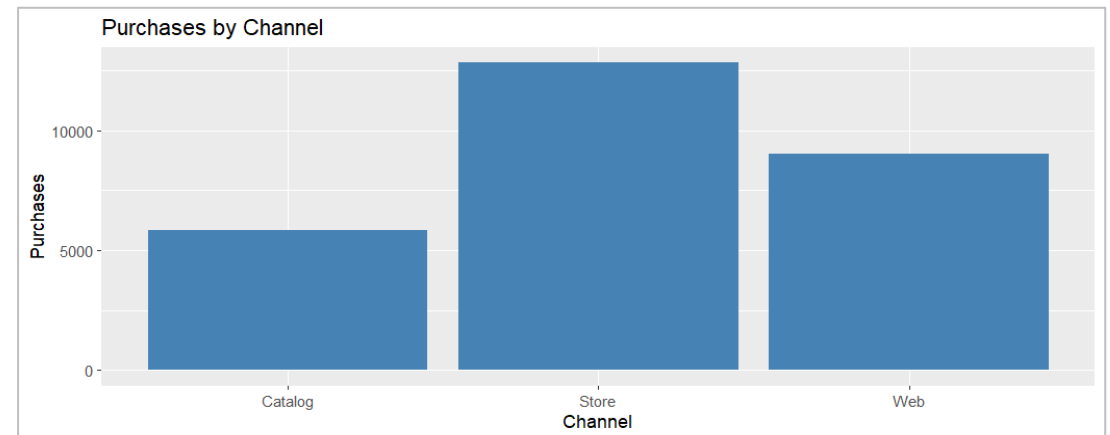
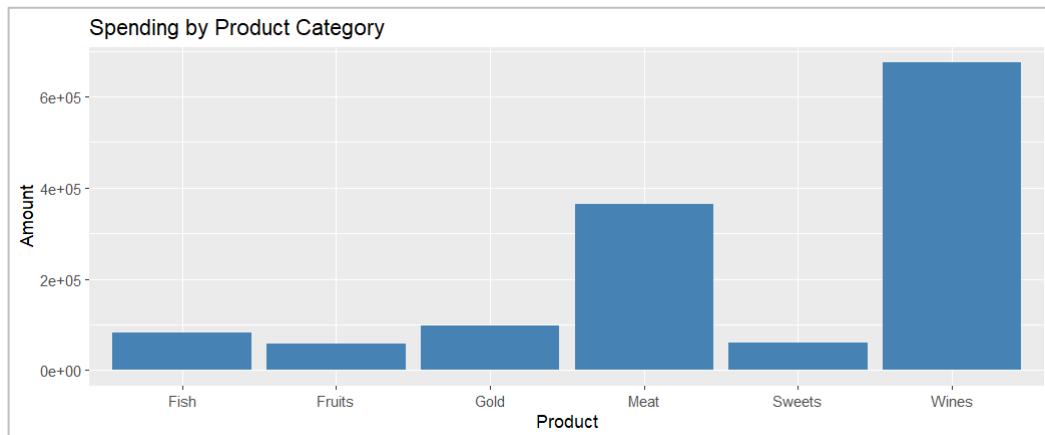
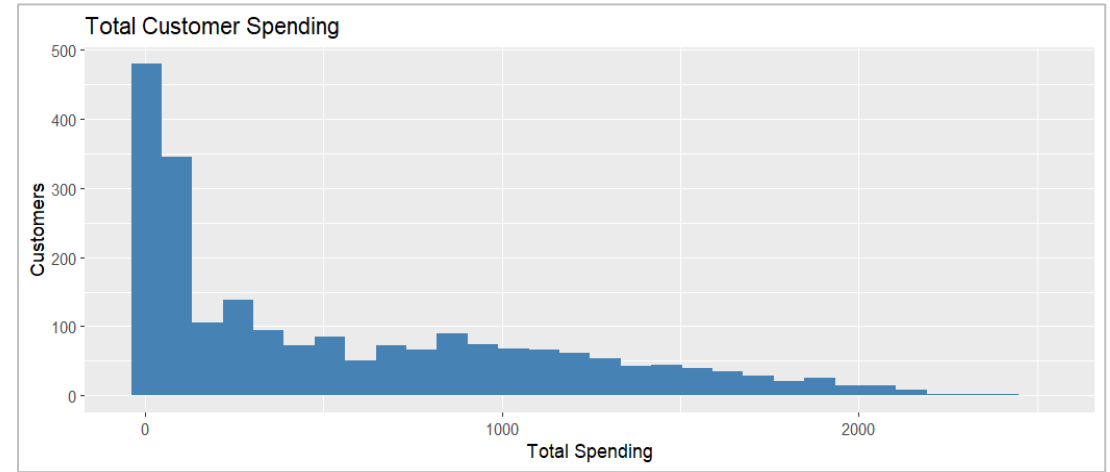
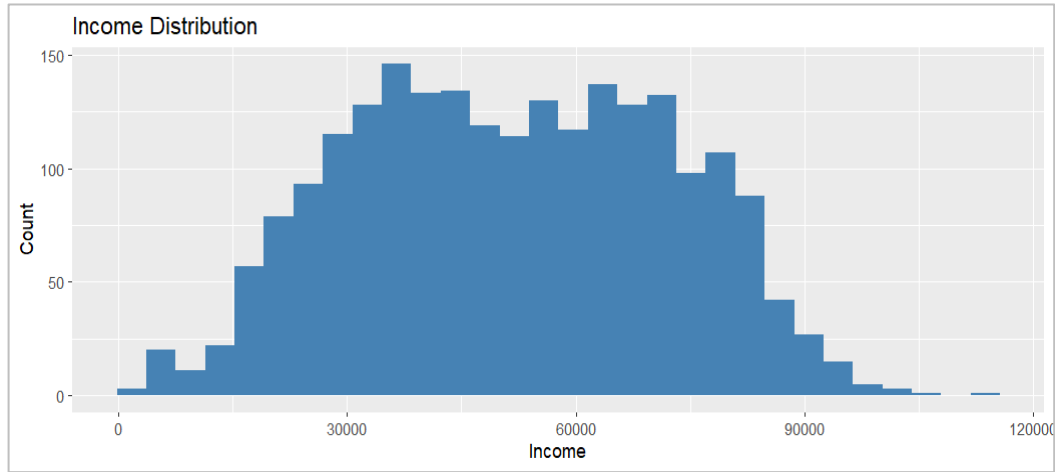
	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response	Age	Customer_Days	marital_Divorced
1	0	0	3	11	1	63	2822	0
2	0	0	3	11	0	66	2272	0
3	0	0	3	11	0	55	2471	0
4	0	0	3	11	0	36	2298	0
5	0	0	3	11	0	39	2320	0
6	0	0	3	11	0	53	2452	0

- constant-value attributes
- provided limited analytical significance for exploratory analysis

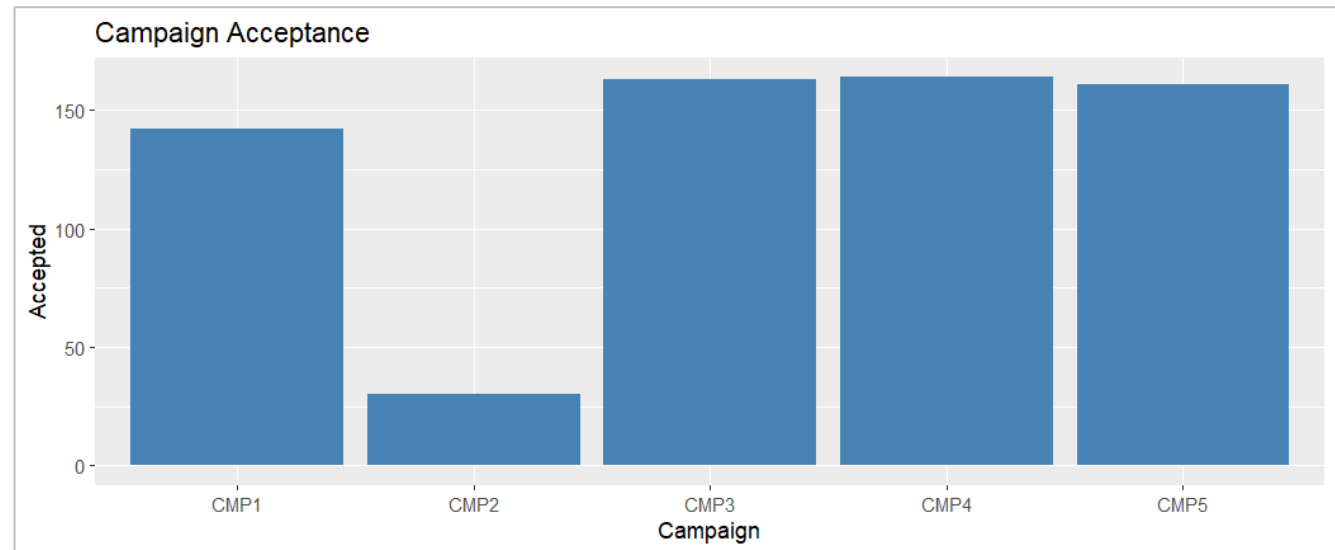
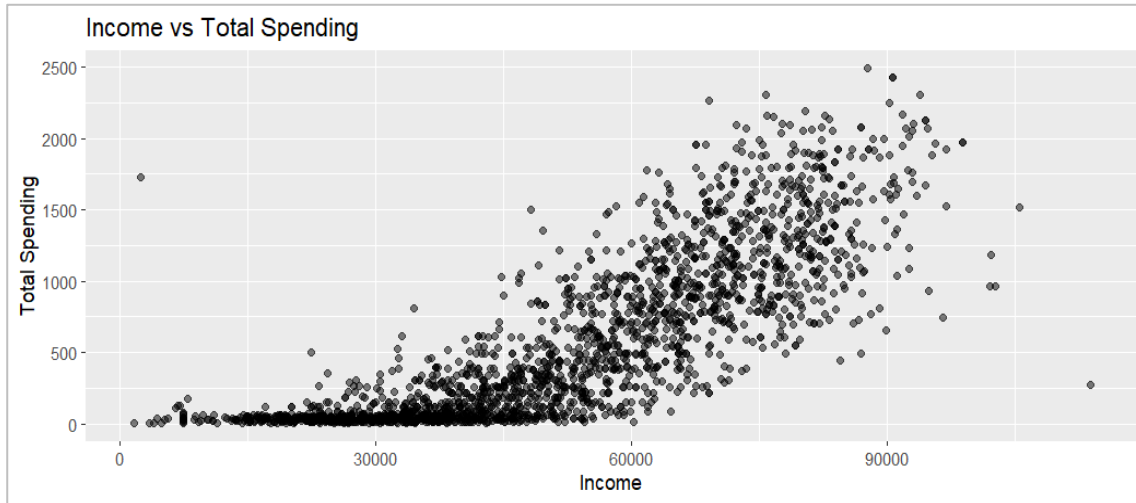
Correlation Between Attributes



Exploratory Analysis



Exploratory Analysis



Dataset Preprocessing

Steps

1. Imported dataset into Power BI
2. Verified and corrected data types
3. Checked missing values (none found)
4. Removed duplicates
5. Detected outliers
6. Removed irrelevant columns (z_cost_contact, z_revenue)
7. Validated existing calculated columns (MntTotal, AcceptedCmpOverall)
8. Created additional calculated columns (TotalChildren, CustomerSegment, PurchaseChannelPreference, AgeGroup, MaritalStatus, Campaign)
9. Renamed columns for dashboard friendliness

Identified and Calculated KPIs

- KPI 1 - Total Customer Spending: Total amount spent by all customers across product categories.
- KPI 2 - Campaign Acceptance Rate: Percentage of customers who accepted at least one marketing campaign.
- KPI 3 - Average Customer Spending: Average spending per customer.
- KPI 4 - Web Purchase Ratio: Percentage of purchases made through the web channel.
- KPI 5 - Customer Complaint Rate: Customer complaint rate.
- KPI 6 - High-Value Customers: For customer segmentation.

Critical Analytical Questions

Question	Importance	Visualization Type	Advanced Features
Which product category generates the highest customer spending?	Helps identify the most profitable product categories and supports inventory and marketing decisions.	Clustered Column Chart	Tooltip for percentage contribution, Slicer for marital status or education
Which customer segment contributes the highest total spending?	Which customer segment spends the most helps businesses prioritize valuable customer groups	Clustered Column Chart	Age group slicer, Drill-through to customer segmentation page
How effective were marketing campaigns?	Measures campaign success and customer responsiveness.	Clustered Bar Chart, Stacked Column Chart	Tooltips with acceptance percentages
Which purchase channel is most preferred by customers?	Helps evaluate customer shopping behavior and optimize channel investments.	Donut Chart	Cross-filtering with customer segments, Slicers for age and income

Critical Analytical Questions

Question	Importance	Visualization Type	Advanced Features
Is customer income related to total spending?	Determines whether higher-income customers spend more.	Scatter Plot	Trend line, Tooltips with customer demographics
Which age groups spend the most and respond better to campaigns?	Supports family-oriented marketing strategies.	Scatter Plot, Stacked Column Chart	Drill-through, Slicers for marital status
How does education level influence customer spending?	Helps identify customer groups most responsive to marketing campaigns.	Clustered Column Chart, Stacked Column Chart	Tooltip with customer counts, Cross-highlighting
Are frequent web visitors making more purchases?	Measures effectiveness of digital engagement.	Scatter Plot	Trend line, Filters for customer segments

Dashboard Demonstration

Key Insights

- Wine is the most spent on product.
- Purchases at the store have the highest value.
- Campaign 3 has the highest acceptance, and campaign 2 has an exceptionally lower acceptance.
- The more the income, the more the spending.
- Graduated persons have the highest spending, and 2n Cycle has the lowest.
- Married persons have the highest spending, and widows have the lowest.
- The most preferred channel is the store

Thank You...!!